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From Prompt to Action: A Comprehensive Review of LLM Autonomous Agents

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***Abstract*—Large Language Models (LLMs) have quickly pushed the frontiers of autonomous agents with advanced reasoning, natural language interaction, and tool chaining in complex worlds. With LLM-based agents arising in domains like digital assistants, autonomous robots, and mission planning, it is more critical than ever to have a deep understanding of their construction, strengths, and weaknesses—especially for safety-critical and adversarial domains like space systems. This article presents an overview of the most recent developments in autonomous agents built with LLMs. We categorize modern architectures, single-agent and multi-agent architectures, and their most prominent functional modules—perception, reasoning, planning, and action. We present new functionality facilitated by LLMs, including zero-shot generalization, dynamic tool use, and human-AI collaboration, and criticize their drawbacks in real-world use, e.g., hallucination, limited resources, and safety. Besides, we discuss future standards and metrics for LLM agents, including how to measure dependability and robustness in hostile environments. Lastly, we present open research challenges highlighting the necessity of stable, efficient, and robust LLM-based agents deployable in wireless, remote, and hostile environments. This survey aims to offer researchers and practitioners a brief overview of the status quo with LLM-based autonomous agents and inspire future work bridging current gaps between general-purpose language intelligence and domain-specific autonomous systems.**

***Index Terms*—Artificial Intelligence, Agent, LLM, GPT-3, GPT-4**

I. Introduction

Autonomous agents have been ubiquitous in artificial intelligence research for a while now, and they find use in all areas of robotics and mission-critical systems, digital assistants, and multi-agent coordination platforms [1]. The recent advances in large language models (LLMs) have added new capabilities to these agents in the form of robust language comprehension, reasoning, and use of tools. LLM-based agents can comprehend complicated commands, decompose goals into actionable steps, communicate with humans in natural language, and use external tools and data sources dynamically [2].

This convergence of LLMs and autonomous agents is enabling an emerging generation of autonomous systems that capitalize on the general-purpose language reasoning capability of foundation models with autonomous decision-making and actuation in the physical world. Such agents are of broad interest in wireless, remote, and hostile environments—e.g., autonomous space mission operation, disaster response, and other applications where robust, flexible, and reliable autonomy is critical [3].

While promising, LLM-based agents are plagued by serious issues that prevent their safe and reliable use in mission-critical settings [4]. Issues such as hallucination, unstable behavior, high computational complexity, limited domain adaptation, etc., call for a better understanding of their strengths and weaknesses [5]. In spite of initial research and industrial proof-of-concepts having produced breathtaking results, a systematic consolidation of this rapidly developing body of research is still required.

We address this gap in this survey by presenting a structured review of autonomous agents based on LLMs. We begin with a description of the key concepts and classification of different agent architectures and frameworks. We then discuss their key capabilities, strengths, and limitations, and review emerging benchmarks and evaluation techniques. We then present directions to guide future research on the creation of reliable, effective, and robust LLM-based agents for complex and adversarial environments.

The rest of this paper is organized as follows: Section II presents background and related work; Section III introduces a taxonomy of LLM-based agent architectures; Section IV discusses their capabilities and emerging applications; Section V analyzes limitations and key challenges; Section VI reviews current benchmarks and evaluation strategies; Section VII outlines future research directions; and Section VIII concludes the paper.

II. Related Work

Large Language Models (LLMs) such as GPT-3, GPT-4, and other transformer-based architectures have given rise to autonomous agents that can reason, plan, and act in complex environments. These agents increasingly integrate external tools, collaborate in multi-agent setups, and adapt through self-reflection. This literature review extends recent developments, detailing architectures, planning strategies, multi-agent systems, benchmarks, ethical concerns, and open research challenges.

Park et al. [6] design generative agents that simulate human-like daily lives in a sandbox game world, showing how LLMs maintain memories, update beliefs, and interact in consistent, believable ways. Their work has inspired research into long-term memory integration for agents. Yao et al. [7] presented

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ReAct, demonstrating that combining reasoning steps with explicit actions significantly improves LLM performance on tasks like HotpotQA and WebShop. Shinn et al. [8] expanded this concept with Reflexion, adding iterative self-critique loops where an agent learns from errors by revisiting previous steps. Nakano et al [9] with WebGPT illustrated practical architectures where an LLM augments its knowledge base with live web browsing, outperforming static closed-book models in factual accuracy.

Similarly, Qin et al. [10] with Toolformer showed that LLMs can be fine-tuned to self-discover when to call APIs, which reduces hallucinations and improves factual correctness. Gao et al. [11] extended this to real-world scenarios through ChatGPT Plugins, connecting language agents to search engines, booking systems, and code execution environments. Schick et al. [12] demonstrated that autonomous tool selection generalizes well across question answering, code completion, and retrieval tasks, strengthening the case for plug-and-play agent tools.

Furthermore, Liang et al. [13] introduced AutoGen, which formalizes multi-agent LLM frameworks where agents communicate, debate, and refine solutions collaboratively. They showed that role specialization (e.g., planner, executor, critic) improves output quality. Du et al. [14] went further with Mind's Eye, where agents run physics simulations to test hypotheses before acting, which marks progress toward embodied, grounded reasoning. Chen et al. [15] highlighted how code generation benchmarks can act as stress tests for autonomous reasoning abilities. Bubeck et al. [16]explored emergent properties of GPT-4, suggesting that LLMs display early signs of reasoning chains and planning capabilities beyond simple next-word prediction. These benchmarks expose both the promise and fragility of LLM-based agents.

Whereas, Bommasani et al. [17] delivered a comprehensive analysis of the risks of scaling general-purpose models into autonomous agents, urging stronger norms for transparency and control. Weidinger et al. [18] systematically mapped potential harms from open-ended deployment, recommending risk assessments at each development stage. Together, these works provide a foundation for ethical guidelines as autonomy grows. Wang et al. [19] with Voyager, showcased a milestone in continuous learning, where an LLM agent explores Minecraft indefinitely, writing its code to acquire new skills and adapt to novel environments, foreshadowing open-ended learning in physical and virtual worlds. Collectively, these studies demonstrate that LLM-based autonomous agents are evolving rapidly but remain constrained by issues in grounding, memory, tool reliability, and alignment. Closing these gaps is crucial for safe, scalable, and beneficial agent deployment in the real world.

III. Taxonomy and Architectures of LLM-Based Agents

Large Language Models (LLMs) such as GPT-3, GPT-4, and related transformer-based models have inspired new forms of autonomous agents that combine general-purpose language reasoning with planning, memory, and tool use in dynamic environments. This section refines our taxonomy of LLM-based agents by explicitly connecting it to recent representative systems described in the literature. We categorize these agents along four key dimensions:

1) Reasoning and Planning Strategies
2) Tool Use and API Integration
3) Memory and Self-Reflection
4) Multi-Agent Collaboration

A. Reasoning and Planning Strategies

Early agent frameworks like ReAct [7]showed that interleaving chain-of-thought reasoning with explicit action steps significantly improves performance on knowledge-intensive tasks. Building on this, Reflexion [8] introduced iterative self-critique loops, where the agent learns from mistakes by revisiting its own reasoning trace. Similarly, WebGPT [9] demonstrated how agents extend static knowledge with dynamic web search to ground their answers more reliably. These systems show that LLM-based agents increasingly rely on explicit reasoning traces, intermediate plans, and dynamic updates rather than single-pass generation. Planning strategies are thus a defining feature of modern LLM architectures.

B. Tool Use and API Integration

One of the most distinctive advances in recent agents is autonomous tool use. Toolformer [12] formalized self-supervised training for LLMs to decide when and how to invoke APIs, reducing hallucinations. ChatGPT Plugins [11] extended this to real-world tasks like web search, booking, and code execution.

These agents follow an architecture where the LLM core acts as a controller that dynamically generates API calls or plugin invocations based on user instructions and intermediate reasoning steps. This plug-and-play pattern is increasingly common and supports applications that require up-to-date external information and real-world actions.

C. Memory and Self-Reflection

Persistent memory and self-reflective loops distinguish advanced LLM agents from simple prompt chains. Park et al. [6] demonstrated generative agents with long-term memory, showing that LLMs can store, retrieve, and update beliefs about an environment or social context. Reflexion [8] complements this with self-critique and iterative learning, where agents refine their outputs by reflecting on errors. Voyager [19] showed that agents can write and adapt their own code to explore open-ended virtual environments, pointing to lifelong learning.

These designs commonly use separate memory modules, logging systems, or vector stores to extend the LLM's context beyond its limited prompt window, allowing it to reason about past states and maintain continuity in long-running tasks.

D. Multi-Agent Collaboration

While many early agents were standalone, new frameworks increasingly explore multi-agent collaboration. AutoGen [13]

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formalized role-based systems where multiple LLMs debate, critique, and refine solutions collaboratively. MetaGPT and ChatDev [10]organize agents into teams with specialized roles (planner, coder, tester, reviewer) to simulate human-like workflows for complex tasks. Mind's Eye [14] extends this by embedding simulation modules that allow agents to test hypotheses before taking actions.

These architectures highlight an important trend: future LLM agents will likely coordinate with other agents or modules, enabling distributed planning, role specialization, and emergent cooperative behaviors. This has significant implications for deploying such agents in multi-agent systems for swarm robotics, remote operations, and autonomous mission planning. Table I presents a clear taxonomy of representative LLM-based autonomous agents, systematically categorizing them by their defining characteristics and use cases.

TABLE I
TAXONOMY OF REPRESENTATIVE LLM-BASED AUTONOMOUS AGENTS

Agent Framework	**Core Idea**	**Key Tools / Actions**	**Notable Capability**	**Benchmark**
ReAct [7]	Reasoning + Acting	Chain-of-Thought + Actions	Improved QA accuracy	HotpotQA, WebShop
Reflexion [8]	Self-critique loop	Iterative self-reflection	Error correction	General NLP tasks
Toolformer [12]	Self-learned tool use	API calling	Reduced hallucinations	QA, Code Completion
AutoGen [13]	Multi-agent conversation	Role-based agents	Collaborative solving	Code generation
Voyager [19]	Continual self-coding	Self-written Python code	Lifelong learning	Minecraft sandbox
MetaGPT [20]	Meta programming	Structured multi-role team	Software pipeline automation	Collaborative coding

IV. CAPABILITIES OF LLM-BASED AUTONOMOUS AGENTS

Large Language Models (LLMs) have significantly expanded what autonomous agents can accomplish by providing flexible, general-purpose reasoning, planning, and natural language interaction. This section highlights the core capabilities that distinguish LLM-based agents from traditional static AI systems and outlines how these abilities enable new forms of autonomy.

A. Natural Language Understanding and Dialogue

LLM-based agents excel at interpreting complex, open-ended natural language instructions and maintaining context-rich multi-turn dialogues with users. Unlike earlier rule-based chatbots, modern agents can parse ambiguous tasks, clarify goals interactively, and adapt instructions to dynamic contexts. This makes them particularly useful for human–AI collaboration, digital assistants, and remote mission operations where human operators provide high-level goals in natural language.

B. Reasoning and Chain-of-Thought

LLMs can perform sophisticated chain-of-thought reasoning, breaking down tasks into sub-goals, checking intermediate steps, and adapting plans on the fly. ReAct [7] showed that explicitly combining reasoning traces with actions improves factual accuracy and task performance, while Reflexion [8]demonstrated how self-critique loops let agents refine outputs by analyzing prior mistakes. This dynamic reasoning makes LLM agents capable of tackling open-domain tasks that static models struggle with.

C. Tool Use and External Knowledge Access

A major advancement is the ability of LLM agents to call external tools, APIs, and plugins to extend their knowledge and act beyond the constraints of their training data. Toolformer [12] showed how LLMs learn when to invoke tools autonomously. ChatGPT Plugins and similar systems integrate LLMs with real-world applications like live web search, booking services, and code execution, enabling practical, goal-directed action. This capability is vital for scenarios where factual correctness and up-to-date information are critical, such as mission planning, disaster response, or autonomous data analysis in remote or extreme environments.

D. Memory and Long-Term Context

A core limitation of vanilla LLMs is their fixed context window. Newer agent designs extend this through persistent memory modules that store and retrieve long-term context. Park et al. [6] demonstrated agents that maintain memories and update beliefs about simulated environments, supporting more realistic, consistent behavior. Reflexion's self-critique loop and Voyager's [19] continuous self-coding expand this to open-ended, lifelong learning.

Memory enables agents to adapt over extended deployments, retain user-specific knowledge, and revisit past decisions, which is especially valuable for autonomous operations where continuity is essential.

E. Autonomous Planning and Execution

Beyond passive conversation, LLM-based agents can autonomously plan sequences of actions and execute them to achieve user-specified goals. For example, WebGPT [9] demonstrated web browsing to gather up-to-date information, while Mind's Eye [14] used embedded simulations to test hypotheses before acting. Voyager's continuous learning agent autonomously explores and codes in Minecraft, acquiring new skills without explicit reprogramming. These planning and execution capabilities make LLM agents suitable for high-autonomy applications in open-world settings, ranging from digital work assistants to physical robots operating in dynamic, uncertain environments.

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F. Multi-Agent Collaboration and Role Specialization

Recent frameworks demonstrate that LLM agents can collaborate in multi-agent setups where each agent assumes a specialized role. AutoGen [13] and MetaGPT [20]show that agents acting as planners, coders, and reviewers can communicate, critique, and refine solutions cooperatively. This supports parallelism, improves solution quality, and mimics human collaborative workflows. Such collaboration has significant implications for large-scale, distributed autonomous systems—like swarms of robots, sensor networks, or distributed mission planning for space or remote operations—where multiple agents must coordinate effectively under uncertainty. Collectively, these capabilities illustrate why LLM-based autonomous agents represent a step change in AI, enabling general-purpose autonomy that adapts, interacts, reasons, learns, and collaborates across diverse tasks and environments.

V. Limitations and Challenges

Despite their impressive capabilities, LLM-based autonomous agents face significant limitations that must be addressed before they can be reliably deployed in safety-critical, mission-critical, or large-scale real-world applications. This section outlines the main technical and ethical challenges that constrain current systems and highlights areas that demand deeper research and robust engineering solutions.

A. Hallucination and Factual Inaccuracy

One of the most persistent issues with LLMs is hallucination, the tendency to generate plausible-sounding but factually incorrect or misleading information. Even when augmented with tools or live retrieval (e.g., WebGPT [9] Toolformer [12] agents can misinterpret retrieved data or fail to verify sources. This undermines trust, especially in contexts where incorrect actions may have serious consequences, such as medical decision support, disaster response, or remote mission operations. Preventing hallucinations requires advances in model alignment, robust fact-checking modules, and better uncertainty estimation. Hybrid architectures that combine LLMs with verifiable symbolic reasoning or external databases may help, but designing such systems remains an open challenge.

B. Computational Cost and Resource Demands

LLMs such as GPT-3 and GPT-4 are resource-intensive to run, especially in long-running or real-time applications. Many current agent frameworks require frequent prompt chaining, external tool calls, or large context windows, which further increases latency and compute overhead. In edge or remote environments—like autonomous drones, space probes, or disaster response robots—hardware limitations make running large models locally impractical. Offloading to the cloud introduces bandwidth constraints, privacy risks, and added latency that may be unacceptable for safety-critical tasks. Research into lightweight LLMs, efficient fine-tuning, distillation, and hardware accelerators is needed to make these agents viable for constrained deployments.

C. Real-Time Responsiveness and Reliability

Many LLM-based agents rely on sequential generation and multiple reasoning-action loops. This can lead to unpredictable execution times and inconsistent response speeds, a major barrier for real-time systems such as autonomous vehicles, collaborative robots, or multi-agent swarms. Unlike traditional control systems with guaranteed timing, LLM-based planners may produce different outputs on repeated runs due to stochastic sampling, introducing nondeterminism that complicates safety validation. Ensuring real-time guarantees, robust fallback mechanisms, and bounded response times is an active area for research and engineering.

D. Safety, Control, and Alignment

As LLM-based agents become more autonomous and capable of real-world actions (e.g., making bookings, executing code, manipulating robots), risks related to unintended or harmful behaviors grow. Bommasani et al. [17] and Weidinger et al. [18] emphasize that foundation models, when scaled into autonomous agents, amplify challenges of misuse, unpredictable emergent behaviors, and unclear responsibility for errors. Current prompt-based controls and post hoc filters are not sufficient for high-stakes settings. There is a need for more robust guardrails, interpretable policies, and human-in-the-loop frameworks to ensure alignment with operator intent and broader societal norms. Ongoing research into constitutional AI, red-teaming, and verifiable agent design will be critical for responsible deployment.

E. Evaluation and Benchmarking Gaps

Existing benchmarks (e.g., WebGPT [9], Codex [15]primarily evaluate static language tasks or narrow tool use. Dynamic agent behavior, especially long-term memory, autonomous goal-setting, and open-ended learning (e.g., Voyager [19])is not well captured by current metrics. New benchmarks are needed to systematically test whether LLM-based agents behave reliably over extended interactions, handle unexpected scenarios, and maintain factual grounding. Developing realistic, domain-specific testbeds for safety-critical domains such as extreme environments, multi-agent swarms, or robotics remains a pressing challenge for the community.

VI. Benchmarks and Evaluation

Robust and standardized evaluation frameworks are essential for understanding the true capabilities and limitations of LLM-based autonomous agents. While significant progress has been made in benchmarking language models on static tasks such as question answering, code generation, and knowledge retrieval, evaluating dynamic agent behavior introduces unique challenges. An overview of representative benchmarks used to gauge the effectiveness of LLM-based agents is provided in Table II.

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TABLE II
REPRESENTATIVE BENCHMARKS FOR EVALUATING LLM-BASED AGENTS

Benchmark	Focus	Measures	Key Limitations
WebGPT [9]	QA with web browsing	Factual correctness	Narrow domain (QA only)
Codex [15] / HumanEval[21]	Code generation	Functional correctness	Limited to programming tasks
Voyager [19]	Lifelong learning	Skill acquisition, self-coding	Game-specific, limited generalization
AutoGen tasks [13]	Multi-agent collaboration	Output coherence, collaboration	Lacks standard metrics for communication
Mind's Eye [14]	Grounded reasoning	Physics simulations, hypothesis testing	Early-stage, limited to simulation

A. Existing Benchmarks for LLM Reasoning and Tool Use

Early benchmarks such as WebGPT [9] illustrated how augmenting LLMs with live browsing can improve factual correctness in open-domain question answering, setting a precedent for evaluating grounded tool use. Similarly, Toolformer [12] demonstrated that LLMs can be fine-tuned to learn when to invoke APIs, with evaluation focused on whether tool use improves factuality and task success across diverse domains. Code generation tasks like Codex [15] and HumanEval [21] have emerged as stress tests for an agent's ability to reason step-by-step, translate instructions into executable programs, and self-correct errors. These benchmarks indirectly measure planning and problem decomposition, which are central to agent autonomy.

VII. FUTURE DIRECTIONS AND OPEN PROBLEMS

The rapid progress in LLM-based autonomous agents opens exciting possibilities for general-purpose, adaptive AI systems that can reason, plan, interact, and learn continuously in open-ended environments. However, realizing this vision requires solving several open research problems and addressing critical deployment risks. This section outlines promising directions for future work.

A. Toward Robust Grounding and Fact-Checking

Reducing hallucination and improving factual reliability remain central challenges. Future agents will likely combine LLMs with external knowledge graphs, symbolic reasoning engines, or verified databases to ground their claims in verifiable facts. Research into uncertainty quantification, better calibration, and automatic fact-checking pipelines is needed to build trust in autonomous reasoning.

B. Memory Architectures and Lifelong Learning

Current memory modules, such as those used in generative agents [6] and Reflexion [8] are still basic approximations of human-like memory. Future work should explore more advanced persistent memory systems that can store rich, structured episodic data, learn to forget irrelevant information, and adapt memories to changing contexts. Continuous or lifelong learning frameworks like Voyager's open-ended self-coding [19] should be extended to real-world physical agents that accumulate skills safely over long horizons.

C. Real-Time, Edge-Compatible LLM Agents

There is a pressing need to adapt LLM-based agents for real-time decision-making in resource-constrained or edge environments. This calls for research on efficient model compression, distillation, and hardware-software co-design for deployment on robots, drones, or remote sensors. Hybrid architectures that offload heavy reasoning to the cloud while keeping core planning and safety modules local may provide a promising balance.

D. Robust Multi-Agent Systems and Emergent Behavior Control

As frameworks like AutoGen [13] and MetaGPT [20] show, multi-agent LLM systems can coordinate, debate, and solve complex tasks cooperatively. However, emergent behaviors in multi-agent systems are hard to predict, verify, and align with human goals. Future work must develop formal guarantees for inter-agent communication, conflict resolution, and fail-safe fallback strategies to prevent undesired outcomes in distributed settings such as swarm robotics or autonomous mission control.

E. New Benchmarks and Standardized Testbeds

To advance the field responsibly, the community needs open, realistic benchmarks for evaluating LLM agents in dynamic, real-world conditions. Sandbox games (e.g., Minecraft for Voyager), simulated robotics labs, and virtual mission control scenarios should be expanded with rigorous metrics for factuality, planning reliability, memory retention, and safety under edge cases. Shared testbeds would encourage reproducibility, robust comparison, and faster progress on hard-to-measure dimensions such as trustworthiness and ethical behavior.

F. Human-AI Collaboration and Ethical Governance

Finally, as LLM agents gain more autonomy and agency, designing effective human-in-the-loop workflows becomes vital to prevent misuse or unintended actions. Future systems should allow humans to steer, inspect, and override agent behavior when necessary. Research into transparent agent policies, auditable decision logs, and explainable reasoning chains will help bridge the gap between black-box models and trustworthy AI assistants. Developing clear governance frameworks and risk assessments, as emphasized by Bommasani et al. [17] and Weidinger et al. [18], remains critical for balancing innovation with safety and societal benefit. Interdisciplinary collaborations will be key to ensuring that the next generation of autonomous agents aligns with human values, especially in high-stakes applications such as exploration, remote operations, or disaster response.

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VIII. Conclusion

Large Language Models (LLMs) have transformed the field of autonomous agents by enabling systems that can reason, plan, interact, and adapt in increasingly open-ended environments. This survey reviewed the state of the art in LLM-based autonomous agents, providing a taxonomy of their architectures, core capabilities, representative frameworks, and practical applications. While recent advances demonstrate impressive progress in natural language understanding, tool use, persistent memory, and multi-agent collaboration, significant challenges remain. Issues such as hallucination, high computational demands, real-time constraints, and the lack of robust safety guarantees highlight the need for continued research into more reliable, transparent, and controllable agent systems. Benchmarking the dynamic behaviors of autonomous agents also remains an open problem, calling for new testbeds and evaluation metrics that capture long-term adaptation, emergent cooperation, and trustworthiness under real-world constraints.

Looking ahead, solving these challenges will require interdisciplinary collaboration across machine learning, control theory, human-computer interaction, and ethics. If guided responsibly, the next generation of LLM-based autonomous agents promises to unlock powerful applications, from digital scientific assistants and collaborative software developers to resilient mission planners in remote or extreme environments, expanding what intelligent systems can do safely and reliably alongside humans.

References

[1] Z. Deng, Y. Guo, C. Han, *et al.*, "Ai agents under threat: A survey of key security challenges and future pathways," vol. 57, no. 7, 2025, ISSN: 0360-0300. DOI: 10.1145/3716628. [Online]. Available: https://doi.org/10.1145/3716628.

[2] L. Hughes, Y. K. Dwivedi, T. Malik, *et al.*, "Ai agents and agentic systems: A multi-expert analysis," *Journal of Computer Information Systems*, vol. 65, no. 4, pp. 489–517, 2025. DOI: 10.1080/08874417.2025.2483832.

[3] A. Chan, C. Ezell, M. Kaufmann, *et al.*, "Visibility into ai agents," in *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT '24, Rio de Janeiro, Brazil: Association for Computing Machinery, 2024, pp. 958–973, ISBN: 9798400704505. DOI: 10.1145/3630106.3658948. [Online]. Available: https://doi.org/10.1145/3630106.3658948.

[4] J. Li, M. Zhang, N. Li, D. Weyns, Z. Jin, and K. Tei, "Generative ai for self-adaptive systems: State of the art and research roadmap," vol. 19, no. 3, 2024, ISSN: 1556-4665. DOI: 10.1145/3686803. [Online]. Available: https://doi.org/10.1145/3686803.

[5] R. W. White, "Advancing the search frontier with ai agents," *Commun. ACM*, vol. 67, no. 9, pp. 54–65, Aug. 2024, ISSN: 0001-0782. DOI: 10.1145/3655615. [Online]. Available: https://doi.org/10.1145/3655615.

[6] J. S. Park and et al., "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023.

[7] S. Yao and et al., "React: Synergizing reasoning and acting in language models," in *NeurIPS 2023*, 2023.

[8] N. Shinn and et al., "Reflexion: Language agents with verbal reinforcement learning," in *ICLR 2023*, 2023.

[9] R. Nakano and et al., "Webgpt: Browser-assisted question-answering with human feedback," in *NeurIPS 2021*, 2021.

[10] C. Qin and et al., "Toolformer: Language models can teach themselves to use tools," in *ICLR 2023*, 2023.

[11] X. Gao and et al., "Chatgpt plugins: Large language models meet apis," in *ACL 2023 Findings*, 2023.

[12] T. Schick and et al., "Toolformer: Language models can teach themselves to use tools," in *ICLR 2023*, 2023.

[13] P. P. Liang and et al., "Autogen: Enabling next-gen llm applications via multi-agent conversation framework," in *ACL 2023 Findings*, 2023.

[14] Y. Du and et al., "Mind's eye: Grounded language model reasoning through simulation," in *NeurIPS 2023*, 2023.

[15] M. Chen and et al., "Evaluating large language models trained on code," in *NeurIPS 2021*, 2021.

[16] S. Bubeck and et al., "Sparks of artificial general intelligence: Early experiments with gpt-4," *Microsoft Research Technical Report*, 2023.

[17] R. Bommasani and et al., "On the opportunities and risks of foundation models," in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 2021.

[18] L. Weidinger and et al., "Ethical and social risks of harm from language models," in *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 2022.

[19] X. Wang and et al., "Voyager: The first lifelong learning llm agent in minecraft," in *ICML 2023*, 2023.

[20] H. Hong, X. Zhu, X. Zhang, *et al.*, "Metagpt: Meta programming for multi-agent collaborative framework," 2023.

[21] F. Zhang, L. Wu, H. Bai, *et al.*, *Humaneval-v: Benchmarking high-level visual reasoning with complex diagrams in coding tasks*, 2025. arXiv: 2410.12381 [cs.CV]. [Online]. Available: https://arxiv.org/abs/2410.12381.

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